Supporting Information for "Core Spacing in Herschel Gould Belt Survey Filaments: a Convention-Limited Sub-Classical Scale and its Interpretation"

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31 July 2026

This document is the online-only Supporting Information for the main paper. It collects the exploratory simulation campaigns and numerical-validation details that are summarised, but not relied upon quantitatively, in the main text. Section, equation, table and figure references of the form "Section X", "Eq. Y" etc. that are not defined here refer to the main paper. The headline fragmentation wavelength rests solely on the near-critical longitudinal growth-rate ladder of the main paper; nothing in this Supporting Information is load-bearing for the main conclusions.

APPENDIX S1: SIMULATION VALIDATION DETAILS

To ensure robustness against numerical effects we performed three validation tests (83 simulations). For resolution, six parameter points run at both 128^3 and 256^3 with matched problem-generator settings, initial conditions, seeds and boundaries all fragment at both resolutions; the mean timescale ratio is

$$\frac{t_{\rm frag}(256^3)}{t_{\rm frag}(128^3)} = 0.915 \pm 0.012, \qquad \text{(S11)}$$

an $8.5\% \pm 1.2\%$ offset (from the ratio 0.915 ± 0.012), all points within a $\pm 11\%$ band (Fig. S11), so the fragmentation classification is resolution-independent. Replacing the Gaussian-profile initial condition with uniform density yields identical classifications at all 10 tested points.

A fragmentation-wavelength measurement is trustworthy only if the local Jeans length remains resolved ($\gtrsim$ 4 cells; Truelove et al. 1997) up to the epoch at which the spacing is measured. The Truelove criterion was derived to prevent *artificial* fragmentation in self-gravitating collapse; we apply it here as a necessary resolution-adequacy condition for a growth-rate measurement, because an under-resolved Jeans length spuriously enhances small-scale power, exactly the grid-scale, resolution-divergent rise we identify as numerical (Fig. ??a). We therefore tracked the minimum cells-per-local-Jeans-length, N_J, along each collapse trajectory (Table S11). For the *longitudinal* runs (both the near-critical calibration and the supercritical ensemble that provides the headline longitudinal value), the inter-bead wavelength ($\lambda \approx \lambda_{\rm J}$, itself resolved by ~ 32 cells) is established at moderate density contrast ($C \approx 20$–40 near-critical; early beading well before runaway for the supercritical ensemble), where $N_J \approx 5$–7 (Truelove satisfied); N_J drops below 4 only at $C > 170$, i.e. *after* the spacing is first set. For the perpendicular *ambipolar* runs, by contrast, beading is measured only at $C \approx 250$–560 where $N_J \approx 1$–2 (Truelove violated), and the short perpendicular values are *resolution-limited upper limits*. This is why we base the quoted wavelength on the linear growth-rate spectrum rather than on late-time peak counts: the growth-rate peak is established when $N_J \gtrsim 5$ and is resolution-converged (the 128^3–512^3 ladder above), whereas the late-time peak count is set by sub-fragmentation at $N_J < 4$ and is not.

Table S11. Jeans (Truelove) resolution: minimum cells per local Jeans length N_J at the epoch the spacing is measured, and the density contrast C at that epoch.

Configuration	C at measurement	min N_J
Near-critical longitudinal (calibration)	20–40	5–7 (OK)
Supercritical longitudinal ensemble (headline)	early beading, moderate C	$\gtrsim 4$ (OK)
Ideal perp., converged ladder (512^3; Section ??)	≈ 3 / ≈ 30	170 / 40 (OK; mor
Ideal perp., short-λ artefact onset	$> 10^2$ (grid-dependent)	$\lesssim 1$ (violatec
Perp. ambipolar (tension side)	250–560	1–2 (violatec

S10.1 Equation of State Sensitivity

Testing mildly sub-isothermal equations of state with $\gamma = 0.9$ and 0.8 at 5 representative points, we find that $\gamma < 1$ systematically accelerates fragmentation (Figure S12). At $\gamma = 0.8$, all 5 test points fragmented at $t_{\rm frag} \approx 0.66$–$0.68\, t_{\rm J}$, approximately half the isothermal fragmentation timescale for the same $(f, \beta, \mathcal{M})$; at $\gamma = 0.9$ fragmentation is delayed (1 of 5 fragments by the integration limit). These are relative comparisons at matched $t_{\rm J}$: the point is that $\gamma < 1$ shortens $t_{\rm frag}$ monotonically, consistent with the isothermal runs of the main text (which fragment at $t_{\rm frag} \approx 1.1$–$1.2\, t_{\rm J}$); we draw no stability inference from the finite integration window.

For $\gamma < 1$ the pressure response to compression weakens ($P \propto \rho^\gamma$ rises more slowly than isothermal), reducing Jeans

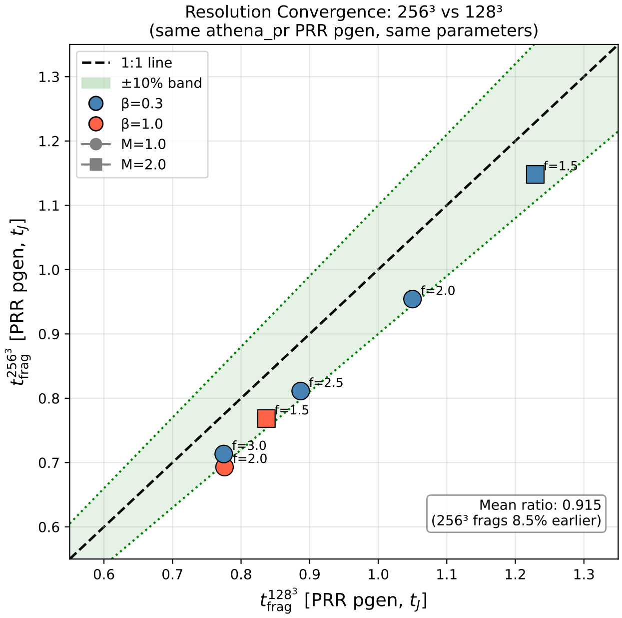

Figure S11. Resolution convergence with matched problem generator: $t_{\rm frag}(256^3)$ versus $t_{\rm frag}(128^3)$ for six parameter points, all within a $\pm 11\%$ band (mean ratio 0.915 ± 0.012).

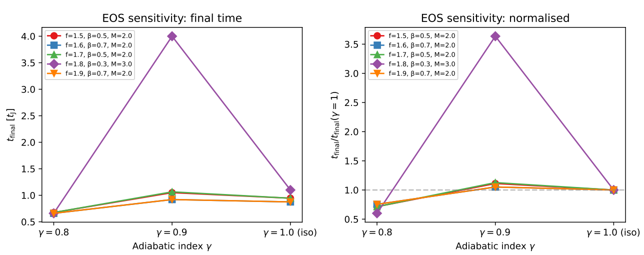

Figure S12. Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of γ (all parameter points). Right: $t_{\rm frag}/t_{\rm final}$ versus γ with lines connecting the same $(f, \beta, \mathcal{M})$ parameter point across γ values. Sub-isothermal EOS ($\gamma < 1$) systematically reduces $t_{\rm frag}$.

support and hastening fragmentation. Far-IR-cooled cloud filaments typically have $\gamma \approx 0.7$–0.95, so the isothermal predictions are conservative: real filaments are if anything more fragmentation-prone.

APPENDIX S2: COMPLETE SIMULATION CAMPAIGN LIST

Table S21 lists every simulation campaign cited in the manuscript. Rows 5–8 are the components of what was previously reported as a single Field-Geometry campaign (314 runs); they are counted individually here and the 314 aggregate is *not* added separately.

APPENDIX S3: SUPPLEMENTARY SIMULATION-CAMPAIGN RESULTS

The campaigns in this appendix are exploratory and are *not* used quantitatively in the main conclusions; the headline wavelength rests solely on the near-critical longitudinal growth-rate ladder of Section **??**.

S31 The radial-collapse/fragmentation timescale competition

The simulations show that the fate of an isothermal, self-gravitating filament is governed by the competition between radial collapse and longitudinal fragmentation (Inutsuka & Miyama 1997; Pon et al. 2012). Near-critical filaments ($f \lesssim 1.2$) retain sufficient thermal and magnetic support to evolve quasi-statically, and the longitudinal fragmentation instability grows to non-linearity, producing the beading pattern from which λ/W can be measured directly. Supercritical filaments ($f \gtrsim 1.5$) fragment longitudinally when properly confined by an ambient medium (Section **??**), at a converged *upper bound* on the wavelength $\lesssim 0.5$ times the classical value (Section **??**). Without ambient confinement the filament's Gaussian skirt feeds an accretion channel through the boundary and suppresses fragmentation. Extended-integration tests confirm that the periodic grid's early termination fires before the longitudinal instability becomes non-linear ($\sigma_{\rm lon}$ onset at $t = 0.42$–$0.71\,t_{\rm J}$ versus a Courant-limited kill at $t \approx 0.25$–$0.34\,t_{\rm J}$), independent of resolution.

In the near-critical regime ($f \approx 1.0$–1.2), filaments fragment with clearly measurable longitudinal density peaks at $t_{\rm frag} \approx 1.1$–$1.2\,t_{\rm J}$ (Section **??**). Runs at the exact Jeans-critical value ($f = 1.00$) likewise bead, at $t_{\rm frag} = 1.62 \pm 0.03\,t_{\rm J}$ across seeds and resolutions (this exact-critical, weak-field subset fragments later than the $f = 1.0$–1.2 field-geometry grid above, whose stronger seeding and field bring it to 1.1–$1.2\,t_{\rm J}$), so there is no stability threshold at the critical line mass; this is the regime in which λ/W is directly measurable.

Supercritical regime ($f \geq 1.5$): Radial collapse dominates. In the periodic supercritical grid ($f = 1.5$–3.0) we find no longitudinal fragmentation structure, only pure radial collapse: the central density increases by two orders of magnitude while the longitudinal density variance remains at $< 0.02\%$ up to and including the fragmentation time. All simulations undergo runaway radial collapse (detected as $\Delta t < 10^{-8}\,t_{\rm J}$ from the CFL limit) before longitudinal beading can develop.

The supercritical radial-collapse time ($t_{\rm coll} \approx 0.25$–$0.34\,t_{\rm J}$) is 3–5 times shorter than the near-critical longitudinal-fragmentation time ($t_{\rm frag} \approx 1.1$–$1.6\,t_{\rm J}$), so in configurations whose initial conditions are out of radial equilibrium the radial mode reaches runaway before longitudinal perturbations grow to non-linearity. The measurable fragmentation wavelengths therefore come from two regimes: the near-critical, quasi-static regime where longitudinal beading is observed directly (Section **??**), and ambient-confined supercritical filaments that fragment longitudinally before radial collapse (Section **??**).

Table S21. Complete list of simulation campaigns underlying the paper.

#	Campaign (section)	Coverage (f
1	Base three-regime grid (§4.5; Fig. S31)	$f \approx 1$; $\beta\, 0.0$
2	Transition grid / DTC (§4.3; Fig. **??**)	$f\, 1.4$–2.2; β
3	Supercritical grid (§4.6)	$f\, 1.1$–3.0; β
4	Validation (§4.4; App. S1)	resolution /
5	Near-critical (§4.7.1)	$f\, 1.0$–1.2; β
6	Perpendicular (§4.7.2)	$\theta = 90^\circ$; $f\, 1$
7	Oblique calibration (§4.7.3)	$\theta = 30/45/6$
8	Adiabatic EOS (§4.7.4; Fig. **??**)	$\gamma = 5/3$; $f\, 1$
9	Turbulence + perp-β + crit.-transition (§4.8; incl. turbulent-support $f_{\rm eff}$ 90 + physical-vs-synthetic turbulence comparison 54)	$\delta v/c_s\, 10^{-4}$–
10	Targeted DTC re-runs (Fig. **??**)	$\beta = 0.3, \mathcal{M}$
11	Direct supercritical ensemble (§4.6.6; Fig. **??**)	$f\, 1.5$–2.0; β
12	Reconciliation suite (§4.6.6)	$f \times \beta \times 2$ se
13	Ideal-MHD perp. $d = 1.0$ grid (§4.6.6)	$\theta = 90^\circ$; $\beta\, 0$
14	Ambipolar non-ideal survey (§4.6.6; Fig. **??**)	$\theta = 90^\circ$; $\eta_{\rm A}$
15	Growth-rate + equilibrium control + transverse/higher-f tests (§**??**, App. **??**)	$f\, 0.7$–2.0; 12
16	Axial-resolution stability (§**??**)	$f = 1.1$; $\theta =$
17	Ideal-perpendicular refinement ladder (§**??**; Table S11)	$\theta = 90^\circ$; $f =$
18	Divergence-free helical grid (§**??**)	$f = 1.1$; B_ϕ
	Athena++ MHD total	

S32 Three fragmentation regimes

The baseline grid, at a near-critical line mass ($f \approx 1$), reveals three fragmentation regimes in the $(\beta, \mathcal{M})$ plane, characterised by the end-of-run density contrast $C = \rho_{\rm max}/\rho_0$ (Fig. S31): (i) *strong-field, magnetically supported* ($\beta \lesssim 0.15$), where magnetic pressure strongly suppresses collapse ($C \approx 1$ at the snapshot time); (ii) *magnetically regulated* ($0.2 \lesssim \beta \lesssim 2$), where fields modify but do not prevent fragmentation ($C = 3$–11); and (iii) *thermally dominated* ($\beta \gtrsim 3$), where fields are too weak to affect the scale ($C = 13$–22). The dispersion relation (Nakamura et al. 1993) predicts near-complete suppression for $\beta \lesssim 0.15$ and $< 10\%$ modification of the most unstable wavelength for $\beta \gtrsim 2$–3, consistent with the empirical boundaries; these boundaries bracket the sampled grid points ($\beta = 0.05, 0.1, 0.15, 0.2, 0.3, \ldots$) at roughly the one-grid-step level rather than being interpolated, so the $\beta \approx 0.15$ and ≈ 3 thresholds carry an uncertainty of $\sim$one grid step. Observed dense-filament conditions ($\mathcal{M} \approx 1$–3, $\beta \approx 0.3$–2; Arzoumanian et al. 2019; Planck Collaboration et al. 2016; Pattle et al. 2017) place most HGBS filaments in the magnetically regulated regime.

S33 Turbulence and critical-transition dependence

Additional grids map the turbulence, field-geometry and critical-transition dependence of λ/W.

S33.1 Turbulence versus λ/W

Comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov) with synthetic perturbations (10^{-4}) at longitudinal field ($f = 1.0$–1.2, $\beta = 0.5, 1, 2$), the fragmentation *wavelength* is unchanged ($< 5\%$; $\lambda/W = 3.44 \pm 0.76$ vs 3.30 ± 0.85; Fig. S32), with a β-dependence in the *raw late-time bead-count* ratio (4.31 ± 0.60 at $\beta = 0.5$ to 2.81 ± 0.13 at $\beta = 2.0$); as the growth-rate analysis shows (Section **??**), this trend is a late-epoch bead-count effect, not a shift in the underlying $\lambda_{\rm max}$, whose β-variation is weak and within the mode-quantisation step. Turbulence does accelerate collapse by $3.2\times$ ($t_{\rm frag} = 0.33 \pm 0.06$ vs $1.06 \pm 0.16\, t_{\rm J}$). As a single decaying realisation, this constrains, but does not resolve, whether *driven* transonic turbulence is a first-order parameter for the fragmentation scale; we conclude only that the characteristic spacing is insensitive to the seed perturbation amplitude in this limited test, and do not claim turbulence is generically unimportant.

S33.2 Perpendicular-field spacing

For perpendicular fields ($\theta = 90^\circ$, $f = 1.2$–1.5, $\beta = 0.3$–2.0; extended $16\lambda_{\rm J}$ periodic axial domain): strong perpendicular B ($\beta \leq 0.5$) gives pure radial collapse with no axial fragmentation, while for $\beta \geq 1.0$ axial beading occurs at $\lambda/W_{\rm core} = 1.25 \pm 0.09$ ($N = 27$), i.e. $\lambda/W_{\rm fil} \approx 0.4$ under the corrected conversion (Eq. **??**), a factor 2–3 *shorter* than the longitudinal-field value at the same β. Field geometry, not β, dominates, and the perpendicular spacing lies far below the observed NN value (fixed-width, periodic-corrected, $\lambda/W \approx 1.9$; raw 2.1).

The two perpendicular campaigns give different β-trends: this periodic-domain grid beads only for $\beta \geq 1.0$, whereas the ambient-confined ideal-MHD grid of Section **??** beads at $\beta = 0.5$ and 2.0 with spindle collapse near $\beta \approx 1$. They differ in transverse confinement and are both numerically unresolved ($N_J \approx 1$–2; Appendix S1), so we draw no physical β-trend from either; a matched-resolution, matched-confinement grid is needed to settle it (Section **??**). The robust statement is only that the perpendicular beading wavelength, where it occurs, is short ($\lambda/W_{\rm fil} \lesssim 0.6$), well below the observed spacing.

S33.3 Critical-transition spacing

Mapping λ/W across $f = 0.9$–1.3 at five β values (Fig. S33) shows no discontinuity at $f = 1.0$ (variation $< 5\%$) and an apparent β-dependence in the *late-time bead count*: $\lambda/W_{\rm core} = 4.74 \pm 0.73, 4.38 \pm 0.59, 3.19 \pm$

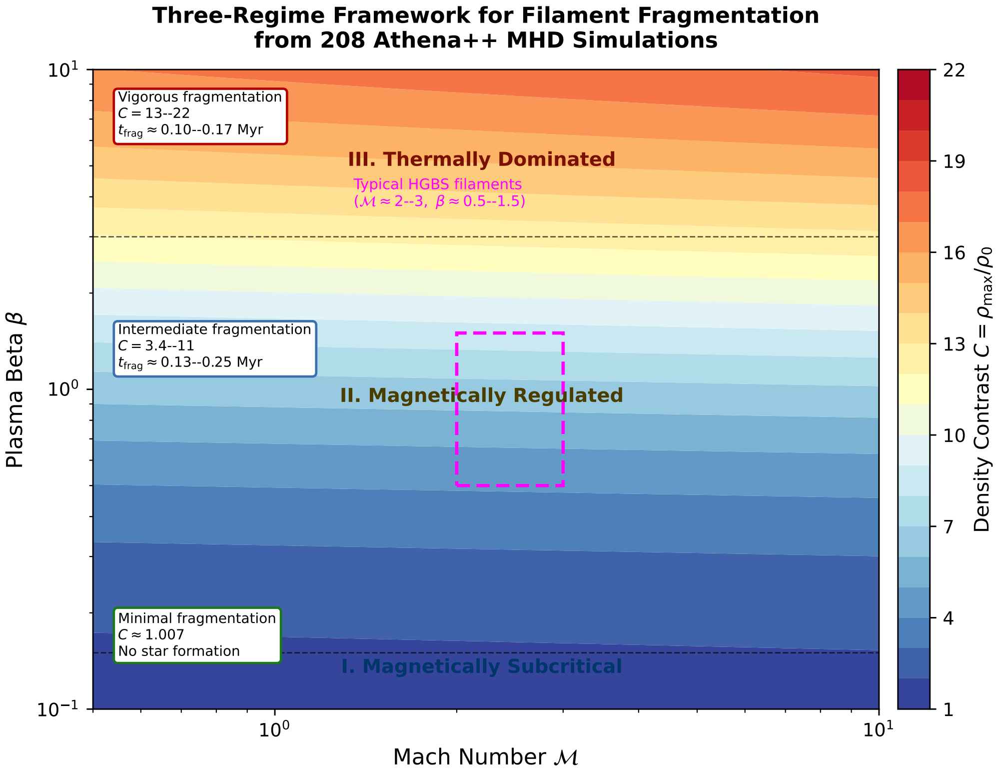

Figure S31. Three-regime fragmentation framework from the Athena++ base grid (near-critical, $\theta = 0°$, isothermal; Section S31). Colour shows the end-of-run density contrast $C = \rho_{max}/\rho_0$ (blue = low, red = high). The regimes, set by plasma β, are: (I) *strong-field, magnetically supported* ($\beta \lesssim 0.15$): magnetic pressure strongly suppresses collapse, $C \approx 1$ (minimal fragmentation at this epoch, not an absence-of-star-formation statement); (II) *magnetically regulated* ($0.2 \lesssim \beta \lesssim 2$): fields modify but do not prevent fragmentation, $C = 3$–11; (III) *thermally dominated* ($\beta \gtrsim 3$): fields too weak to affect the scale, $C = 13$–22. Typical HGBS filament conditions ($\mathcal{M} \approx 1$–3, $\beta \approx 0.3$–2; Arzoumanian et al. 2019; Planck Collaboration et al. 2016; Pattle et al. 2017) lie in the magnetically regulated regime. The regime here is set by the *dynamical* suppression of collapse (a β criterion) and is physically distinct from the magneto-static mass-to-flux criticality of Table **??** (Section S31); the two are not in tension.

0.29, 2.86 ± 0.18, 2.80 ± 0.14 at $\beta = 0.3, 0.5, 1.0, 1.5, 2.0$. This bead-count β-trend is, however, larger than the primary growth-rate measurement: repeating the growth-rate-spectrum method (Section **??**) across $\beta = 0.3, 0.5, 1.0$ gives $\lambda_{max} = 1.00$, 1.14, $0.80\,\lambda_J$, i.e. order one Jeans length with only a weak, mode-quantised β-dependence. We therefore treat the β-dependence of the wavelength as weak (the stronger bead-count trend being a late-epoch, resolution-sensitive artefact) and the two highest-β bead-count points ($\lambda/\lambda_J \approx 0.84$) as consistent with the supercritical ensemble; the observable λ/W_{fil} then follows from the end-to-end forward model (Section **??**) rather than the shortcut factor. **Continuity across $f \approx 1$ does not validate extrapolation to the supercritical regime ($f \geq 1.5$)**, where the mode changes to radial collapse (Section **??**); the near-critical λ/W values are a hypothesis for supercritical filaments, not a measurement.

S33.4 Synthesis

Table S32 summarises the simulation λ/W measurements and their role in the comparison with observation.

With the exception of the 54-run physical-versus-synthetic turbulence *comparison* (Section S33; distinct from the 90-run turbulent-*support* f_{eff} grid of Table S31, both components of the 289-run aggregate campaign 9 of Table S21), the grid uses near-laminar seeding ($\delta v = \mathcal{M} c_s \times 10^{-4}$) rather than driven ISM turbulence. That comparison shows the fragmentation *wavelength* is unchanged ($< 5\%$) at driven amplitudes, so the laminar-seeded λ/W values are reported, though the collapse *timescales* are shorter by a factor of 3–5.

In convergence-independent units the two first-order parameters are field geometry (perpendicular $\lambda/W_{core} \approx 1.25$ versus longitudinal 2.8–4.4) and, for longitudinal fields, plasma β; the observable comparison uses the growth-rate wavelength (Section **??**), not these raw ratios.

Table S31. Effective line-mass fraction with turbulent support.

Quantity	Value	Source
Nominal f (thermal)	1.5 ± 0.3	HGBS line-mass
Turbulent support f_{eff}/f	0.82 ± 0.11	TURB (90 sims)
Effective f_{eff}	1.23 ± 0.28	product
HGBS ensemble range	0.7–1.9	propagated

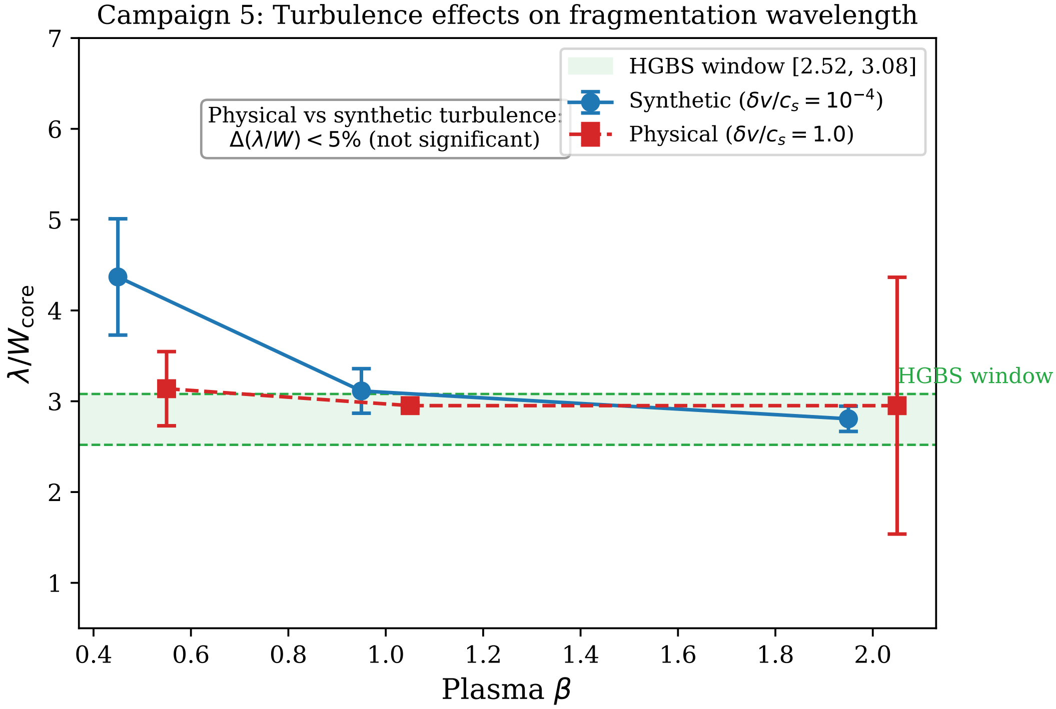

Figure S32. turbulence comparison: physical vs synthetic turbulence give statistically indistinguishable $\lambda/W_{\rm core}$ ($\Delta\mu < 5\%$, KS $p = 0.43$). The β-trend in the raw late-time bead-count ratio ($\sim$4.3 at $\beta = 0.5$ to $\sim$2.8 at $\beta = 2.0$) is preserved across the two turbulence prescriptions, but is a late-epoch bead-count effect rather than a shift in the underlying growth-rate $\lambda_{\max}$, whose β-variation is weak (Section **??**).

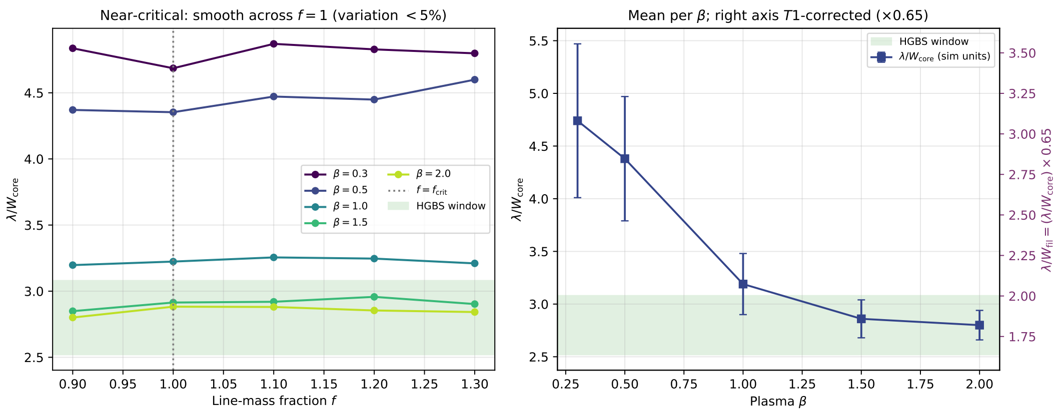

Figure S33. Near-critical runs: $\lambda/W_{\rm core}$ across $f = 0.9$–1.3 for five β values, smooth with no discontinuity at $f = 1.0$ (left); mean per β (right). The convergence-independent quantity is $\lambda/W_{\rm core}$; the observable follows from $\lambda/W_{\rm fil} \approx 0.27\,(\lambda/W_{\rm core})$ (Eq. **??**), so the highest-β points correspond to $\lambda/W_{\rm fil} \approx 0.75$–$0.8$. This near-critical calibration lies below the observed spacing under any convention (Table **??**); the primary simulated value is the growth-rate wavelength (an upper limit; Section **??**), not this ratio.

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Table S32. Simulation fragmentation scale by configuration, in the two convention-independent forms: the raw internal ratio $\lambda/W_{\rm core}$ (normalised by the initial half-width) and the background-density Jeans-length ratio $\lambda/\lambda_{\rm J} = 0.3\,(\lambda/W_{\rm core})$. We do *not* tabulate the observable $\lambda/W_{\rm fil}$ here, since it depends on the width convention; the comparison with observation uses the growth-rate wavelength expressed in *matched* beam-convolved Plummer units, $\lambda/W \approx 1.0$–1.3 (≈ 0.35–0.5 times the classical value, a factor ≈ 2–3; Section **??**), which is set against the observed $\lambda/W \approx 0.4$ (FilFinder) to 1.9 (fixed-width DisPerSE).

Configuration	$\lambda/W_{\rm core}$	$\lambda/\lambda_{\rm J}$	Status
Longitudinal field (quantitative)			
Long. near-critical (β=2.0, calibration)	2.80±0.14	0.84	Calibration only
Long. supercritical ensemble (§**??**)	3.27	0.98	Near one Jeans length; see growth-rate result
Perpendicular field — PRELIMINARY, numerically unconverged (N_J≈1–2); not for quantitative use			
Perp. near-critical ($\beta\geq$1, ideal-MHD)	[1.25]	[0.38]	unconverged
Perp. ambipolar ($\eta_{\rm AD}$=0.001–0.1)	[$\sim$2.2]	[0.66]	unconverged

Only the longitudinal rows are used quantitatively, and even there the reference value is the convergently-bounded growth-rate wavelength (an upper limit; Section **??**, Table **??**), not these raw ratios. The perpendicular values are bracketed to signal that they are numerically unconverged ($N_J \approx 1$–2, below the Truelove criterion; wavelengths shift by factors 2–5 between resolutions) and must not be read as measurements.